

Lab 16: Population & Systems Ecology (Module 15)

BIOL-1

Name: _____ Date: _____

Overview

You are the **acting manager** of the fictional **Green Mesa National Wildlife Refuge**. Elk were reintroduced in 2010; your field team sends **population notes** and **energy budgets** (all invented but numerically consistent). You will **forecast** growth, **diagnose** crashes, and **audit** energy through a food web—like a real reserve plan, but on one sheet of paper.

Learning Objectives

By the end of this lab, you should be able to:

1. Interpret **exponential vs logistic** growth from verbal and tabular data.
2. Classify **density-dependent** vs **density-independent** regulation.
3. Assign **trophic roles** and trace **energy** and **nutrients** through an ecosystem.
4. Apply the **~10% energy transfer** rule across multiple steps.

Materials Needed

- Pencil or pen
- This worksheet
- Calculator (optional)

Part 1: Elk Come Home — From Boom to Ceiling

Elk counts (made-up):

Year Adult elk Notes

2012	40	Few predators; abundant browse
2014	90	Still rising fast

Year Adult elk Notes

2016	155	Growth slowing
2018	198	Near stable
2020	205	Fluctuates ± 5

1. Shape of growth: Which curve best fits 2012–2018—**exponential (J)** or **logistic (S)**? **Circle one** and cite **two** observations from the table that support your choice.

2. Carrying capacity: From the table, estimate **K** (a range is fine, e.g., “about ____ elk”). What **resource** is most likely limiting elk here if the story continued?

- Estimated K:

- Limiting resource (hypothesis):

3. Prediction: If managers **remove half the elk** in 2021 and browse recovers, what do you predict happens to population growth rate the next few years—**increase / decrease / stay same**? Why?

Part 2: Two Crashes — Which Regulator?

Two bad years, two mechanisms.

Event A: A **parasitic fly** spreads; infection probability rises as elk crowd into the same valleys.

Event B: A **wildfire** burns 40% of the refuge regardless of elk density.

4. Classify each as density-dependent or density-independent.

Event	Type
Event A	
Event B	

Event	Type

5. Management puzzle: Which event could managers partly reduce by **moving elk** or **thinning the herd**? Which is mostly “weather and luck”?

- Herd management helps more with:

- Mostly luck:

6. Why are apex predators numerically rare **even when prey are common?** (energy argument, 2–3 sentences)

Bridge: **Trophic levels** connect **who limits whom** (Part 2) to **how much energy** reaches the top of the food web (Part 4).

Part 3: Web Assembly — Who Eats Whom?

Refuge food web (simplified):

- **Sunlight** → bunchgrass & shrubs (**producers**)
- **Grasshoppers** eat plants
- **Ground-feeding birds** eat grasshoppers
- **Hawks** eat birds
- **Coyotes** eat elk calves (sometimes) and scavenge
- **Fungi & bacteria** decompose dead material

7. Build one linear chain with four trophic levels (producer → ... → top carnivore). Use arrows **from food to eater**. (Example format—your chain can differ: grass → grasshopper → ground-feeding bird → hawk.)

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8. Label roles for the chain you wrote (producer, primary consumer, secondary, tertiary):

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9. Nutrients vs energy: In 2–3 sentences, explain why **decomposers** matter to **producers** even though energy does **not** cycle from dead matter back to sunlight.

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Part 4: Energy Audit — The 10% Rule

Use ~10% transfer between trophic levels (textbook-style rounding). Each step, about one-tenth of the energy in the previous level is available at the next—round answers to a sensible number of digits.

Given: Plants fix **1,000,000 kcal** of chemical energy in tissues eaten by herbivores in a season.

10. Step the energy down the chain:

Step	Available energy (kcal)
To primary consumers (herbivores)	
To secondary consumers	
To tertiary consumers	

11. Sanity check: If humans tried to feed the world only on **apex tuna**, would we need **more** or **less** ocean primary production than if we ate **plants** directly? Explain using the table logic.

12. One-sentence contrast:

- Energy through ecosystems:

- Nutrients through ecosystems: